Irreducible compositions in an exceptional unicritical family

Abstract

We construct irreducible compositions in a positive-shift subfamily left exceptional by a recent theorem on unicritically generated semigroups. Let $d \geq 4$ be even and let $a \geq 2$ be an integer which is not a pth power for any odd prime $p \mid d$, nor a fourth power when $4 \mid d$. Put $f(x) = x^d + a - a^d$ and $g(x) = x^d + a$. We prove that $f \circ g \circ g \circ h$ is irreducible over $\mathbb{Q}$ for every composition h of integral unicritical polynomials whose degrees have prime divisors among those of d. For a finite equal-degree generating set containing f and g, this gives an explicit positive lower density. Within the exceptional four-constant alphabet, the shorter prefix $f \circ g$ suffices. The result includes infinitely many pairs in every degree divisible by four for which g is reducible and f fixes a square. The proof combines one real embedding, a field norm, and gaps between consecutive prime powers.

1 Introduction and statements

Let $\mathcal{A}$ be a finite set of polynomials of degree at least two in $\mathbb{Z}[x]$, and let $G = \langle \mathcal{A} \rangle$ be the semigroup they generate under composition. A natural question is whether the presence of one irreducible polynomial forces a positive proportion of the elements of G to be irreducible over $\mathbb{Q}$, when polynomials are counted by degree. For generators of common degree d, write

$$\underline{\delta}(G) = \liminf_{B \to \infty} \frac{\#\{F \in G : \deg F \leq B,\ F \text{ irreducible over } \mathbb{Q}\}}{\#\{F \in G : \deg F \leq B\}}.$$

The prime-degree case was treated in [2], following the quadratic work [3]. For general common degree, Bhardwaj, Boyer-Paulet, Hindes, Qiu, and Sun [1, Theorem 1.1] establish the desired positive-proportion conclusion outside explicit exceptional families. In even degree, the exceptional constants are contained in

$$\{y^p - y^{pd},\ y^p,\ -y^p,\ -y^p - y^{pd}\}, \qquad p \mid d \text{ prime}, \quad y \in \mathbb{Z}. \tag{1}$$

They expect the positive-proportion conclusion to extend to the exceptional cases. We treat a positive-shift subfamily of (1).

Throughout, superscripts on integers denote ordinary powers. We write compositions explicitly, including $g \circ g$ rather than g^2, to distinguish iteration from multiplication. The empty composition is the identity polynomial.

Theorem 1.1. *Let $d \geq 4$ be even, and let $a \geq 2$ be an integer satisfying*

1. *$a \notin \mathbb{Z}^p$ for every odd prime $p \mid d$;*
2. *if $4 \mid d$, then $a \notin \mathbb{Z}^4$.*

Define

$$D = a^d - a, \qquad f(x) = x^d - D, \qquad g(x) = x^d + a, \qquad P = f \circ g \circ g.$$

Then $P \circ h$ is irreducible over $\mathbb{Q}$ for every finite composition h of polynomials $x^m + c$ with $c \in \mathbb{Z}$, $m \geq 2$, and $\operatorname{rad}(m) \mid \operatorname{rad}(d)$. This includes the empty composition.

Corollary 1.2. *Under the hypotheses of Theorem 1.1, let G be generated by r distinct polynomials $x^d + c_i$, with $c_i \in \mathbb{Z}$, including f and g. Then*

$$\underline{\delta}(G) \geq r^{-3}.$$

If instead the generating set is a subset of

$$\{x^d + a - a^d,\ x^d + a,\ x^d - a,\ x^d - a - a^d\} \tag{2}$$

containing f and g, then $f \circ g \circ h$ is irreducible for every $h \in G$ and for the empty composition, and

$$\underline{\delta}(G) \geq r^{-2}.$$

In particular, $\underline{\delta}(\langle f, g\rangle) \geq 1/4$.

The main new instances occur when $4 \mid d$ and $a = 4z^4$. If z is odd and nonzero, the hypotheses hold for every such d: the valuation $v_2(a) = 2$ excludes every odd perfect power and every fourth power. For $d = 2^k$, $k \geq 2$, every nonzero integer z is allowed. In these instances $a = (2z^2)^2$ and $f(a) = a$, while

$$g(x) = \left(x^{d/2} - 2zx^{d/4} + 2z^2\right)\left(x^{d/2} + 2zx^{d/4} + 2z^2\right) \tag{3}$$

is reducible. Thus an irreducible positive-shift generator is not available to obtain the conclusion by the existing norm argument. The first example is

$$f = x^4 - 252, \qquad g = x^4 + 4.$$

The two-letter prefix $f \circ g$ gives the lower bound $1/4$ in the semigroup generated by this pair.

The argument in [1, Proposition 3.8] excludes the positive-shift partner in (2). Its norm sufficient condition [1, Proposition 3.1] gives no conclusion at the first transition here, since $f(g(0)) = a$ is a square in the displayed subfamily. Our proof deals with that transition using a negative real root of f and then separates the fourth-power obstruction by its norm. All later transitions follow from elementary gaps between prime powers. We do not resolve the full exceptional locus (1).

2 Algebraic irreducibility criteria

We use the classical binomial criterion: for a field K of characteristic zero and $n \geq 2$, the polynomial $x^n - b$ is irreducible over K if and only if

$$b \notin K^p \quad (p \mid n \text{ prime}), \qquad b \notin -4K^4 \quad \text{if } 4 \mid n. \tag{4}$$

See [4, Chapter VI, Section 9, Theorem 9.1]. The exceptional fourth-power clause in (4) is essential in composite even degree.

Lemma 2.1 (Capelli's composition criterion)**.** *Let $A \in K[x]$ be irreducible, let α be a root of A, and let $B \in K[x]$ be nonconstant. Then $A \circ B$ is irreducible over K if and only if $B(x) - \alpha$ is irreducible over $K(\alpha)$.*

Proof. If β is a root of $B(x) - \alpha$, then $\alpha = B(\beta)$ and $K(\alpha) \subseteq K(\beta)$. The tower law gives

$$[K(\beta) : K] = [K(\beta) : K(\alpha)] \deg A.$$

The right side equals $(\deg B)(\deg A)$ precisely when $B(x) - \alpha$ is irreducible over $K(\alpha)$. Since β is a root of $A \circ B$, the equality is equivalent to irreducibility of that polynomial. $\square$

The following familiar norm consequence is included to specify the parity and fourth-power issue exactly; compare [1, Proposition 3.1].

Lemma 2.2. *Let $A \in \mathbb{Q}[x]$ be monic and irreducible of even degree N, and let $B = x^m + c$ with $m \geq 2$ and $c \in \mathbb{Q}$. If $A(c) \notin \mathbb{Q}^p$ for every prime $p \mid m$, then $A \circ B$ is irreducible over $\mathbb{Q}$.*

Proof. Suppose otherwise, and put $K = \mathbb{Q}(\alpha)$ for a root α of A. Lemma 2.1 and (4) imply either $\alpha - c = u^p$ for a prime $p \mid m$, or $4 \mid m$ and $\alpha - c = -4u^4$, with $u \in K$. In the first case its norm is a rational pth power. In the second case its norm is

$$(-4)^N \mathrm{N}_{K/\mathbb{Q}}(u)^4,$$

which is a rational square because N is even. In both cases this contradicts the assumptions, since

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha - c) = (-1)^N A(c) = A(c).$$

□

3 The first mixed composition

Lemma 3.1. *Let $d \geq 4$ be even and $p \mid d$ be prime. If $b \geq 2$ is an integer, then*

$$(b^{d/p})^p - (b^{d/p} - 1)^p > b^{d-d/p} \geq b^2.$$

Proof. For an integer $u \geq 2$, factor the difference of powers to obtain

$$u^p - (u-1)^p = \sum_{j=0}^{p-1} u^{p-1-j}(u-1)^j > u^{p-1}.$$

Apply this with $u = b^{d/p}$. Since $d \geq 4$ is even, $d - d/p \geq 2$. □

Proposition 3.2. *Under the hypotheses of Theorem 1.1, both f and $f \circ g$ are irreducible over $\mathbb{Q}$.*

Proof. For every prime $p \mid d$, Lemma 3.1 gives

$$(a^{d/p} - 1)^p < a^d - a = D < (a^{d/p})^p.$$

Thus $D \notin \mathbb{Z}^p$, hence $D \notin \mathbb{Q}^p$. Here we use the elementary fact that an integer which is a rational pth power is an integer pth power. Since $D > 0$, it also does not belong to $-4\mathbb{Q}^4$. Criterion (4) proves f irreducible.

Choose its negative real root $\alpha = -D^{1/d}$ and regard $K = \mathbb{Q}(\alpha)$ as a subfield of $\mathbb{R}$. The element $\alpha - a$ is negative in this embedding, so it is not a square in K. Also

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha - a) = f(a) = a. \tag{5}$$

For an odd prime $p \mid d$, equation (5) and the hypothesis $a \notin \mathbb{Z}^p$ exclude $\alpha - a \in K^p$.

It remains to address the exceptional clause when $4 \mid d$. If $\alpha - a = -4u^4$ for some $u \in K$, then, since $[K : \mathbb{Q}] = d$,

$$a = (-4)^d \mathrm{N}_{K/\mathbb{Q}}(u)^4 = \left(4^{d/4} \mathrm{N}_{K/\mathbb{Q}}(u)\right)^4.$$

This would make a a rational fourth power, and therefore an integer fourth power, contrary to hypothesis. Thus (4) proves $g(x) - \alpha = x^d - (\alpha - a)$ irreducible over K. Lemma 2.1 proves $f \circ g$ irreducible over $\mathbb{Q}$. □

4 Uniform propagation

Proof of Theorem 1.1. If $y \geq a^d + a$ is an integer, then $D < y$. For every prime $p \mid d$, Lemma 3.1 therefore gives

$$(y^{d/p} - 1)^p < f(y) = y^d - D < (y^{d/p})^p. \tag{6}$$

In particular, $f(y)$ is not a rational pth power for any prime $p \mid d$.

Put $H = f \circ g$, irreducible by Proposition 3.2. Since $g(a) = a^d + a$, equation (6) applies to $H(a)$. Lemma 2.2 now proves $P = H \circ g$ irreducible.

For every $t \in \mathbb{Z}$, evenness of d implies $g(t) = t^d + a \geq a$, and hence

$$g(g(t)) \geq a^d + a.$$

Equation (6) yields

$$P(t) \notin \mathbb{Q}^p \qquad (t \in \mathbb{Z},\ p \mid d \text{ prime}). \tag{7}$$

We can therefore append the factors of h one at a time. Suppose that $Q = P \circ h_0$ is already irreducible, and append $B = x^m + c$ as in the theorem. Then $h_0(c) \in \mathbb{Z}$, so

$$Q(c) = P(h_0(c)) \notin \mathbb{Q}^p \qquad (p \mid m \text{ prime})$$

by (7). The polynomial Q remains monic and has even degree. Lemma 2.2 proves $Q \circ B$ irreducible and completes the induction. □

Proposition 4.1. *Under the hypotheses of Corollary 1.2 for the alphabet (2), $f \circ g \circ h$ is irreducible for every finite word h in that alphabet, including the empty word.*

Proof. Put $T = \{t \in \mathbb{Z} : |t| \geq a\}$. All four generator constants belong to T. The first three maps in (2) take T into integers at least a, because $a^d - a \geq a$. For the last map, an input of magnitude a maps to $-a$. For all other inputs in T, integrality gives $|t| \geq a + 1$, and

$$(a+1)^d - a^d \geq da^{d-1} > 2a.$$

Thus its output is greater than a. Consequently every word in the alphabet preserves T.

For $t \in T$ we have $g(t) \geq a^d + a$, so (6) implies that $H(t) = (f \circ g)(t)$ is not a rational pth power for any prime $p \mid d$. Starting from H irreducible by Proposition 3.2, append letters as before. Each required norm value has the form $H(h_0(c))$, where the generator constant c belongs to T and $h_0(c) \in T$. Lemma 2.2 proves irreducibility at every step. □

5 Counting by degree

We include the elementary same-degree freeness argument to pass from words to actual polynomials; compare [1, Proposition 3.9].

Lemma 5.1. *In characteristic zero, distinct polynomials $x^d + c_1, \ldots, x^d + c_r$, with common degree $d \geq 2$, generate a free semigroup under composition.*

Proof. Degree determines word length. Suppose two words of equal length greater than one define the same polynomial. Removing their outer letters leaves monic polynomials U, V of equal positive degree M, with

$$U^d + b = V^d + c.$$

If $U \neq V$, then

$$\deg(U^d - V^d) = (d-1)M + \deg(U - V) > 0,$$

since $(U^d - V^d)/(U - V)$ has leading coefficient d and degree $(d-1)M$. This contradicts $U^d - V^d = c - b$. Hence $U = V$ and $b = c$. Induction recovers all letters. $\square$

Proof of Corollary 1.2. By Lemma 5.1, there are exactly r^n distinct polynomials of word length n, each of degree d^n. Every word of length $n \geq 3$ with prefix fgg is irreducible by Theorem 1.1; there are r^{n-3} such words. Therefore, for $N = \lfloor \log_d B \rfloor \geq 3$,

$$\frac{\#\{F \in G : \deg F \leq B,\ F \text{ irreducible}\}}{\#\{F \in G : \deg F \leq B\}} \geq \frac{\sum_{n=3}^{N} r^{n-3}}{\sum_{n=1}^{N} r^n}.$$

Since $f \neq g$, we have $r \geq 2$, and the right side tends to r^{-3}. For the restricted alphabet, Proposition 4.1 supplies the prefix fg of length two, giving the analogous limit r^{-2}. $\square$

Remark 5.2. The restrictions on a in Theorem 1.1 are sufficient hypotheses for the real-place and norm proof; no necessity assertion is intended. The result does not settle exceptional pairs with the negative shift $x^d - a$ in place of g, when the positive-shift generator is absent. The mixed-degree assertion concerns irreducibility of the indicated compositions, not a density theorem for a mixed-degree semigroup.

References

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